WORKED SOLUTION

STEP III 1997, Question 5

Pure • Cycloid Arc Length

STEP III 1997, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/97-s3-q5

The key idea

A point on the rim of a rolling wheel traces a *cycloid*. Once we write down where that point is in terms of the angle the wheel has turned through, the question is just an arc-length integral. The rolling condition (no slipping) is what links the centre's motion to the angle, and a half-angle identity makes the square root in the arc-length formula collapse.

Solution

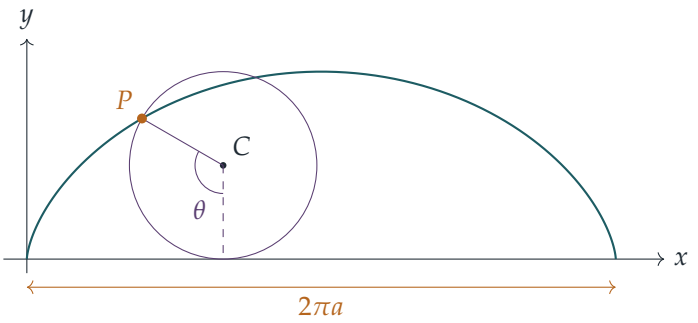
Setting up the path

Let the wheel have radius a and roll without slipping along the x-axis. Start with the marked point P touching the ground at the origin, and let θ be the angle (in radians) through which the wheel has turned.

Because there is no slipping, the length of ground covered equals the length of rim that has touched it, so the centre C has moved a distance $a\theta$ and sits at $(a\theta,\, a)$. Relative to C, the point P started directly below it and has turned through θ (clockwise, as the wheel rolls to the right), so P is at $C + (-a\sin\theta,\, -a\cos\theta)$. Hence

$$x = a(\theta - \sin\theta), \qquad y = a(1 - \cos\theta), \qquad 0 \le \theta \le 2\pi,$$

for one revolution. In that time the wheel moves forward a distance $2\pi a$.



The arc length

Differentiating,

$$\frac{\mathrm{d}x}{\mathrm{d}\theta} = a(1-\cos\theta), \qquad \frac{\mathrm{d}y}{\mathrm{d}\theta} = a\sin\theta,$$

so

$$\begin{aligned}\left(\frac{\mathrm{d}x}{\mathrm{d}\theta}\right)^2 + \left(\frac{\mathrm{d}y}{\mathrm{d}\theta}\right)^2 &= a^2\left(1 - 2\cos\theta + \cos^2\theta + \sin^2\theta\right)\\ &= 2a^2(1-\cos\theta)\\ &= 4a^2\sin^2\tfrac{\theta}{2},\end{aligned}$$

using $1-\cos\theta = 2\sin^2\frac{\theta}{2}$.

> **Ask yourself** When we take the square root, is it $2a\sin\frac{\theta}{2}$ or $2a\left|\sin\frac{\theta}{2}\right|$, and does it matter here?

The square root is always $2a\left|\sin\frac{\theta}{2}\right|$. For $0 \le \theta \le 2\pi$ we have $0 \le \frac{\theta}{2} \le \pi$, where $\sin\frac{\theta}{2} \ge 0$, so the modulus can be dropped over exactly one revolution. (Over a longer range we would have to split the integral.)

The distance travelled by P is therefore

$$s = \int_0^{2\pi} 2a\sin\tfrac{\theta}{2}\,\mathrm{d}\theta = \Big[-4a\cos\tfrac{\theta}{2}\Big]_0^{2\pi} = -4a(-1) + 4a = 8a.$$

The ratio

The wheel moves $2\pi a$ while the point on its rim travels $8a$, so the required ratio is

$$\frac{2\pi a}{8a} = \frac{\pi}{4} \approx 0.785.$$

It doesn't depend on the size of the wheel, as it shouldn't: scaling the wheel scales both distances by the same factor.

Checks

- The point travels further than the wheel ($8a > 2\pi a \approx 6.28a$), which makes sense: it moves forward *and* goes up and down by $2a$ each way. In fact $8a$ is exactly four diameters.
- A crude bound: the path goes from $(0,0)$ up to $(\pi a, 2a)$ and down to $(2\pi a, 0)$, so it is at least $2\sqrt{\pi^2 a^2 + 4a^2} \approx 7.45a$, and at most $2\pi a + 4a \approx 10.28a$. The value $8a$ sits comfortably between.
- Numerically, following the rim point at 200 000 positions as the wheel rolls and adding up the straight-line steps gives 7.99999999992 for $a = 1$, so the ratio is $0.78539816\ldots = \pi/4$.

Common slips

- Getting the signs in the parametrisation wrong (for example $x = a(\theta + \sin\theta)$), which gives a path that moves backwards relative to the wheel. Check it at $\theta = \pi$: P should be at the top, $(\pi a, 2a)$.
- Leaving the integrand as $a\sqrt{2 - 2\cos\theta}$ and trying to integrate it directly; the half-angle identity is the step that makes it easy.
- Writing the ratio upside down: the question asks for the wheel's distance over the point's distance, which is $\pi/4$, not $4/\pi$.

Beyond STEP

The cycloid is one of the most celebrated curves in mathematics. It is the *brachistochrone*, the curve of fastest descent under gravity between two points, and the *tautochrone*, along which a bead reaches the bottom in the same time wherever it starts. Both facts are proved with the calculus of variations, which you meet in a first- or second-year course on mechanics or methods; this question's arc-length calculation is the first step in each. For more problems in this spirit, try Siklos, *Advanced Problems in Mathematics*.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.